

## Assignment-1

Scholar No. 3

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1) Explain dropout with example.

Dropout is a technique to prevent overfitting in a neural network by dropping some neurons during training.

This forces the network to -

- Not depend too much on any single neuron.
  - Learn more robust and generalization patterns.
- It is used only during training and not during testing.

Eg:-

Suppose we have this small network for images classification.

Input (784)  $\rightarrow$  Hidden Layer (6 neurons)  $\rightarrow$  Output (1 neuron)

Hidden Layer activations:-

$$H = [h_1, h_2, h_3, h_4, h_5, h_6]$$

Apply dropout = 0.5 (50%)

During training, randomly 3 neurons are dropped.

$$H = [h_1, x, h_3, x, h_5, x]$$

Next batch different neurons drop.

$$H = [x, h_2, x, h_4, x, h_6]$$

For one training step:

$$y = w_1 \times h_1 + w_3 \times h_3 + w_5 \times h_5 + b$$



Next step:

$$y = w_2 \times h_2 + w_4 \times h_4 + w_6 \times h_6 + b$$

So, the network cannot rely on any single neuron better generalization!

2) Explain Pruning with example.

Pruning means removing less important neurons or weights from a trained neural network to make it smaller and faster.

Advantage:- Reduce model size.

- Reduce computation.

- Speed up inference.

- Save memory.

Eg: Consider this trained network.

Input  $\rightarrow$  Hidden (5 neuron)  $\rightarrow$  Output.

After training, suppose the weights look like this -

$$w_1 \rightarrow 0.85$$

$$w_2 \rightarrow 0.02$$

$$w_3 \rightarrow 0.91$$

$$w_4 \rightarrow 0.01$$

$$w_5 \rightarrow 0.78$$

In pruning step, we set a threshold:

If weight  $< 0.05$ , then remove it.

So, we remove

$$w_2 = 0.02$$

$$w_4 = 0.01$$

After pruning, network becomes

Input  $\rightarrow$  Hidden (3 neuron)  $\rightarrow$  Output.



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# Sigmoid : It is a very common activation function used in AI and neural network.

- It takes any real no. as input and squeeze it b/w 0 & 1. That's why it's often used when we want an o/p that's probabilistic.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Uses :

i) Introduces non-linearity?

Without activation  $f(x)$  neural n/w would act like simple learn model.

ii) Probability interpretation?

Since o/p is b/w 0 & 1, it's perfect for -  
(a) binary & logistic regression.

Limitation :

i) Vanishing Gradient Problem.

ii) For very large & small i/p gradient becomes tiny, training becomes slow.



# Softmax : i) Softmax is an activation function mainly used in the o/p layers of a neural n/w for multi-class classification problem.

ii) It converts a list of numbers into ~~possible~~ probabilities such that -

a) All values are b/w 0 & 1.

b) Sum of all o/p = 1

$$Z_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}} \quad \text{where } z_i = \text{i/p score of classification}$$

$n = \text{no. of class}$

Uses : i) Given prob. distribution

ii) Helps in multi-class classification.

iii) Makes comparison b/w class easy.

iv) Works perfectly with cross entropy.

Limitation : i) Sensitive to large values.

ii) Used only in o/p layers not hidden layers.

# Max out : It is an activation function used in neural n/w that outputs the max value from a set of linear functions.

$$\text{Maxout}(x) = \max(w_1 x_1 + b_1, w_2 x_2 + b_2, \dots, w_k x_k + b_k)$$

where  $k$  = no. of linear function.  
 $w$  = weight & bias.

Uses : No vanishing gradient problem can approximate any act. f<sup>n</sup> works very well with dropout.  
 More flexible than ReLU.

Assignment-3

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Q- Consider 2 components G1 &amp; G2 ...

$$\mu_1 = 2, \mu_2 = 6$$

$$\sigma_1^2 = 6, \sigma_2^2 = 4$$

$$\pi_1 = 0.6, \pi_2 = 0.4$$

Apply GMM on 1D &amp; 2D sample.

• GMM in 1D:

in 1D, the probability density function is -

$$P(x) = \pi_1 N(x|\mu_1, \sigma_1^2) + \pi_2 N(x|\mu_2, \sigma_2^2)$$

$$P(x) = 0.6 N(x|2, 6) + 0.4 N(x|6, 4)$$

when

$$N(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Let,  $x=4$ 

component 1,

$$N(4|2, 6) = \frac{1}{\sqrt{12\pi}} e^{-\frac{(4-2)^2}{12}} = 0.092$$

weighted comp.,  $0.6 \times 0.092 = 0.0552$ .

component 2,

$$N(4|6, 4) = 0.092$$

weighted comp.,  $0.4 \times 0.092 = 0.0368$ .

component 2,

$$N(4|6, 4) = 0.121$$



weighted comp. -  $0.4 \times 0.121 = 0.0484$

Final GMM is

$$P(4) = 0.0552 + 0.0484 \\ = 0.1036$$

Now,

GMM in 2D -

$$u_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad u_2 = \begin{bmatrix} 6 \\ 6 \end{bmatrix} \quad \because \text{we have no info, we assume both to same.}$$

$$\Sigma_1 = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}, \quad \Sigma_2 = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

2D Gaussian PDF is given by -

$$N(u, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|} \exp\left(-\frac{1}{2} (x-u)^T \Sigma^{-1} (x-u)\right)$$

let's say pt.  $x = (4, 4)$

so comp 1,

$$\Sigma_1^{-1} = \begin{bmatrix} 1/6 & 0 \\ 0 & 1/6 \end{bmatrix} \quad (x-u_1) = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad = 0.666$$

$$N_1(4,4) = \frac{1}{2\pi \sqrt{36}} e^{-0.666} = 0.021$$

wt. comp.,  $0.6 \times 0.021 = 0.0126$

Comp 2,

$$\Sigma_2^{-1} = \begin{bmatrix} 1/4 & 0 \\ 0 & 1/4 \end{bmatrix} \quad (x-u_2) = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

$$\frac{1}{2} (4/4 + 4/4) = 1$$

$$N_2(4,4) = 0.0146$$

wt. comp.,  $0.4 \times 0.0146 = 0.0058$

Final 2D GMM is  $\rightarrow P(4,4) = 0.0126 + 0.0058 = \underline{\underline{0.0184}}$